

FPGA Based Home Safety and Automation System

A real-time home safety and automation system designed on the **Xilinx Spartan-7 (XC7S50-CSGA324)** FPGA using **Verilog HDL**. The system integrates 6 sensors and a central alarm to monitor gas leaks, fire, air quality, water level, temperature/humidity, and secure entry – all through a single FPGA controller.

Final Year B.Tech Project – Department of Electronics & Communication Engineering, Babasaheb Bhimrao Ambedkar University, Lucknow (2022–2026)

Overview

Modern homes need to be both safe and comfortable. This project uses the parallel processing power of an FPGA (instead of a sequential microcontroller) to monitor multiple hazards simultaneously and respond in real time – without any dependency on internet or external network connectivity.

Features

-  **Flame Detection** – IR flame sensor triggers alarm + display on fire detection
-  **Gas Leak Detection** – MQ-2 sensor monitors for combustible gas leaks
-  **Air Quality Monitoring** – MQ-135 sensor detects poor indoor air quality
-  **Water Level Monitoring** – Auto-activates motor on low level, alarms on overflow
-  **Temperature & Humidity Monitoring** – DHT11 sensor with range-based LED indication
-  **Password-Based Door Lock** – Servo-controlled lock, opens only on password match
-  **Central Alarm System** – Unified buzzer alert triggered by any hazard condition
-  **Seven-Segment Display** – Real-time status messages for each subsystem

Tech Stack

Category	Tools/Components
FPGA Board	AMD/Xilinx Spartan-7 (XC7S50-CSGA324)
HDL	Verilog
Design Software	Xilinx Vivado

Category	Tools/Components
Schematic Design	KiCad
Sensors	MQ-2 (gas), MQ-135 (air quality), DHT11 (temp/humidity), IR flame sensor, water level sensor
Actuators	Servo motor (door lock), active buzzer (alarm)

System Architecture

All sensors and components connect to the FPGA through GPIO pins:

Component	FPGA Pin
Flame Sensor	C17
Gas Sensor (MQ-2)	B18
Air Quality Sensor (MQ-135)	R7
Water Level Sensor	C14
Servo (Door Lock)	J4 / S0
DHT11 (Temp/Humidity)	L4

Component	FPGA Pin
Buzzer (via LM358 op-amp)	C13

Repository Structure

```

├── home_automation_top.v      # Top-level
    module + all sub-modules
├── home_automation_top_tb.v   # Testbench
    for simulation
└── README.md

```

Simulation

The design was verified using Xilinx Vivado's simulator. Each sub-module (gas detection, flame detection, water level, air quality, door lock, temperature/humidity) was individually tested with waveform analysis to confirm correct input-output behavior before full system integration.

To simulate:

1. Open Vivado → create a new RTL project
2. Add `home_automation_top.v` as a design source
3. Add `home_automation_top_tb.v` as a simulation source
4. Run Behavioral Simulation

Performance Summary

Tested across multiple FPGA families for comparison:

FPGA Series	Total Power (W)	LUTs	Slices	Delay (ns)
Spartan-7 (used)	0.109	238	138	4.948
Artix-7	0.086	239	141	3.391
Zynq-7000	0.107	239	134	4.762
Kintex-7	0.13	239	134	4.762

Resource utilization on Spartan-7: ~1% LUTs/FFs, 30% I/O – indicating an interface-heavy, lightweight design well within the device's capacity.

Future Work

- pH sensor integration for water quality monitoring
- GSM/SMS alerts with battery backup for power-failure resilience
- RTC module for event timestamping
- Fault detection for sensor malfunction

- ASIC implementation for large-scale deployment

Team

- Nisha Tiwari
- Kaveri Maurya
- Shriya Samanta
- Abhinav Kumar

Supervisor: Dr. Mangal Deep Gupta, Assistant
Professor, Dept. of ECE, BBAU Lucknow

License

This project was developed for academic purposes as part of a B.Tech final year thesis.